Macrophage Reverse Cholesterol Transport in Mice Expressing ApoA-I Milano

Eric T. Alexander, Ginny L. Weibel, Michelle R. Joshi, Charulatha Vedhachalam, Margarita de la Llera-Moya, George H. Rothblat, Michael C. Phillips, Daniel J. Rader

Objective—To compare the abilities of human wild-type apoA-I (WT apoA-I) and human apoA-I_{Milano} (apoA-I_M) to promote macrophage reverse cholesterol transport (RCT) in apoA-I–null mice infected with adeno-associated virus (AAV) expressing either WT apoA-I or apoA-I_M.

Methods and Results—WT apoA-I– or apoA-I_M–expressing mice were intraperitoneally injected with [H^3]cholesterol-labeled J774 mouse macrophages. After 48 hours, no significant difference was detected in the amount of cholesterol removed from the macrophages and deposited in the feces via the RCT pathway between the WT apoA-I and apoA-I_M groups. Analysis of the individual components of the RCT pathway demonstrated that the apoA-I_M–expressing mice promoted ATP-binding cassette transporter A1 (ABCA1)-mediated cholesterol efflux as efficiently as WT apoA-I but that apoA-I_M had a reduced ability to promote cholesterol esterification via lecithin cholesterol-acyltransferase (LCAT). This resulted in reduced cholesteryl ester (CE) and increased free cholesterol (FC) levels in the plasma of mice expressing apoA-I_M compared to WT apoA-I. These differences did not affect the rate of delivery of labeled cholesterol to the liver via SR-BI–mediated selective uptake or its subsequent excretion in the feces.

Conclusion—Within the limits of the in vivo assay, WT apoA-I and apoA-I_M are equally efficient at promoting macrophage RCT, suggesting that if apoA-I_M is more atheroprotective than WT apoA-I it is not attributable to an enhancement of macrophage RCT. **(*Arterioscler Thromb Vasc Biol*. 2009;29:1496-1501.)**

Key Words: apoA-I Milano ■ reverse cholesterol transport ■ high-density lipoprotein ■ macrophage ■ apolipoprotein

Apolipoprotein A-I (apoA-I) is the major protein component of HDL.[1,2] Plasma apoA-I and HDL cholesterol levels are inversely associated with the risk of cardiovascular disease.[3] The antiatherogenic properties of apoA-I are thought to arise, at least in part, from its central role in reverse cholesterol transport (RCT), the pathway by which excess cholesterol is effluxed from peripheral tissues and transported to the liver, where it is subsequently excreted from the body.[3–5]

The first natural mutant of apoA-I identified was the apoA-I_{Milano} (apo A-I_M) mutation (R173C).[6] The apoA-I_M mutation is located in the N-terminal helix bundle domain of the protein.[7,8] Naturally occurring variants of apoA-I between residues 121 and 186 are often associated with low apoA-I levels.[9] Individuals heterozygous for the apoA-I_M mutation have very low plasma apoA-I and HDL cholesterol levels as well as moderately elevated triglycerides.[10] Despite a lipid profile that is usually associated with a high risk of premature cardiovascular disease, apoA-I_M carriers display no increase in cardiovascular disease or events.[10–12] This has led to speculation that apoA-I_M is a gain-of-function mutation that has enhanced cardio-protective effects,[13–19] whereas others believe that wild-type (WT) apoA-I and apoA-I_M are functionally equivalent.[20,21] A clinical trial of repeated intravenous infusions of apoA-I_M–phospholipid complexes demonstrated regression of existing atheromas after 5 weekly treatments,[22,23] whereas other clinical trials focused on the ability of WT apoA-I/phosphatidylcholine disks to increase endothelial function.[24,25]

To date it has not been determined how the R173C substitution affects RCT in vivo. To investigate this question we performed the first head-to-head comparison of in vivo macrophage-RCT between WT apoA-I and apoA-I_{Milano} using an assay developed in our laboratory.

Methods

Preparation of ApoA-I Mutants

The cDNAs of interest were expressed in *Escherichia coli* and the resulting proteins purified to greater than 95% purity by gel filtration chromatography.[7] All proteins were stored at −80°C in lyophilized form and before use were dissolved in the appropriate buffer containing 6 mol/L Gdn HCl and dialyzed extensively before use.

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From the Lipid Research Group (E.T.A., G.L.W., M.R.J., C.V., G.H.R., M.C.P.), the Children's Hospital of Philadelphia, and the Department of Medicine (D.J.R.), University of Pennsylvania School of Medicine, Philadelphia.

Correspondence to Dr Daniel J. Rader, University of Pennsylvania Medical Center, 654 Biomedical Research Building II/III, 421 Curie Blvd, Philadelphia, PA 19104. E-mail rader@mail.med.upenn.edu

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Table. Plasma Lipid and ApoA-I Levels

	RCT 1§			RCT 2†		
	WT	Milano	LacZ	WT	Milano	LacZ
Total cholesterol‡	104±13	43±14*	21±3*	36±2	43±7	31±4
HDL cholesterol	80±13	31±2*	14±1*	24±4	30±5	17±3*
Triglycerides	41±5	61±22*	31±4	29±9	35±8	25±10
Phospholipids	199±23	139±11*	69±11*	100±12	106±18	84±13
ApoA-I	154±4	121±9*	0	33±9	125±10*	0

§ApoA-I–null mice injected with adeno-associated virus expressing either WT apoA-I (1×10^{12} GC), apoA-I$_{Milano}$ (1×10^{12} GC), or LacZ (1×10^{12} GC). n=6 per group.

†ApoA-I–null mice injected with adeno-associated virus expressing either WT apoA-I (2×10^{11} GC), apoA-I$_{Milano}$ (1×10^{12} GC), or LacZ (1×10^{12} GC). n=6 per group.

‡All values expressed as mean±SD in mg/dL.

*$P<0.05$ compared to WT.

Preparation of ApoA-I Adeno-Associated Virus

WT apoA-I cDNA was mutated using the Quikchange Site-Directed Mutagenesis Kit from Stratagene. The resulting cDNA were sequenced to confirm the presence of the intended mutation and submitted to the University of Pennsylvania Vector Core for use in creating the liver-specific apoA-I adeno-associated virus (AAV) serotype 8.[26]

Macrophage RCT Studies

Experiments were performed in male apoA-I–null mice obtained from Jackson Labs and fed a chow diet. For each experiment, 18 mice (n=6/group) received i.p. injection of AAV8 (1×10^{12} GC) containing either WT apoA-I, apoA-I$_M$, or LacZ cDNA. On day 42 after vector injection, each animal received intraperitoneal injections of [^{3}H]cholesterol-labeled J774 cells. Blood was collected at 2, 6, 24, and 48 hours. Feces was collected from 0 to 48 hours, and after exsanguinations at 48 hours bile and liver samples were collected as previously described.[27]

Endogenous LCAT Assays

The endogenous LCAT cholesterol esterification rate (CER) in whole plasma was measured using a modified Stokke and Norum procedure.[28–30] Seventy-five microliters of fresh nonradioactive plasma were incubated with a BSA solution containing 3×10^6 dpm of [^{3}H]cholesterol on ice overnight to allow the [^{3}H]cholesterol tracer to equilibrate evenly across the entire spectrum of lipoproteins. The plasma was then incubated at 37°C for 30 minutes as duplicates while a third aliquot was maintained at 4°C as a control. Lipids were extracted by TLC. Endogenous LCAT activity was expressed as nanomoles of CE formed per mL of plasma during the 30 minutes incubation.

Cholesterol Efflux

After [^{3}H]cholesterol labeling the cells (J774 or mouse peritoneal macrophage [MPM]), media containing the appropriate acceptor was added for up to 24 hours. In some experiments the ACAT inhibitor CP113 818 (2 μg/mL) was added to the media. To determine the cholesterol efflux, media were sampled at indicated times, filtered, and counted by liquid scintillation counting to determine the [^{3}H] released. [^{3}H] in the media was compared with total [^{3}H] at time zero to determine the percent release of [^{3}H]cholesterol.[31–33] ABCA1-mediated efflux was calculated by $\%efflux_{+cAMP} - \%efflux_{-cAMP}$.

Cholesterol Influx

Rat Fu5AH hepatoma cells were prepared as described previously[34] and incubated with 20% serum containing [^{3}H]cholesterol and [^{3}H]cholesteryl ester from mice expressing either WT apoA-I or apoA-I$_M$. After 8 hours, the cells were washed 3 times with PBS, the cell lipids were extracted with isopropyl alcohol as previously described,[35] and the levels of tritium label determined.

Data Analysis

Data are from representative experiments and are expressed as mean±SD. Statistical test for significance was done using an unpaired *t* test or 1-way Anova followed by a Tukey test for pairwise comparisons.

Additional materials and methods are available in the supplemental materials (available online at http://atvb.ahahournals.org).

Results

In Vivo RCT Studies

ApoA-I–null mice (n=6 per group) were IP injected with AAV-expressing WT apoA-I, apoA-I$_M$, or LacZ control at a dose of 1×10^{12} genome copies (GC). Six weeks after infection, WT apoA-I– and apoA-I$_M$–expressing mice roughly reproduced the relative apoA-I and lipid levels of the human apoA-I$_M$ carriers and their controls[36]; apoA-I$_M$–expressing mice had modestly lower plasma apoA-I and substantially lower HDL-C levels compared with WT apoA-I-expressing mice (Table). Differences between the groups were not attributable to differential gene expression, as hepatic mRNA levels of WT apoA-I and apoA-I$_M$ were similar (supplemental Figure I). At week 6 after AAV injection, cholesterol-labeled J774 cells were injected for a macrophage RCT study. The [^{3}H]-cholesterol counts in plasma, expressed as a percentage of total labeled cholesterol injected, at each time point were significantly higher for WT apoA-I compared to apoA-I$_M$ or LacZ mice (Figure 1A). Expression of human WT apoA-I significantly increased the transport of macrophage-derived cholesterol into the feces compared to the LacZ control (Figure 1B). However, there was no significant difference in counts in either the liver and bile (supplemental Table I) or the feces (Figure 1B) between WT apoA-I and apoA-I$_M$ mice despite the lower plasma levels of apoA-I and HDL cholesterol for apoA-I$_M$–expressing mice. FPLC profiles of pooled plasma samples from the 48-hour time point indicated that the HDL formed by WT apoA-I and apoA-I$_{Milano}$–expressing mice were similar (supplemental Figure II).

In a subsequent experiment the dose of WT apoA-I AAV was reduced 5-fold, whereas the apoA-I$_M$ dose was held constant in an attempt to equalize the level of HDL-C between WT apoA-I– and apoA-I$_M$–expressing mice. This treatment resulted in substantially reduced apoA-I levels in

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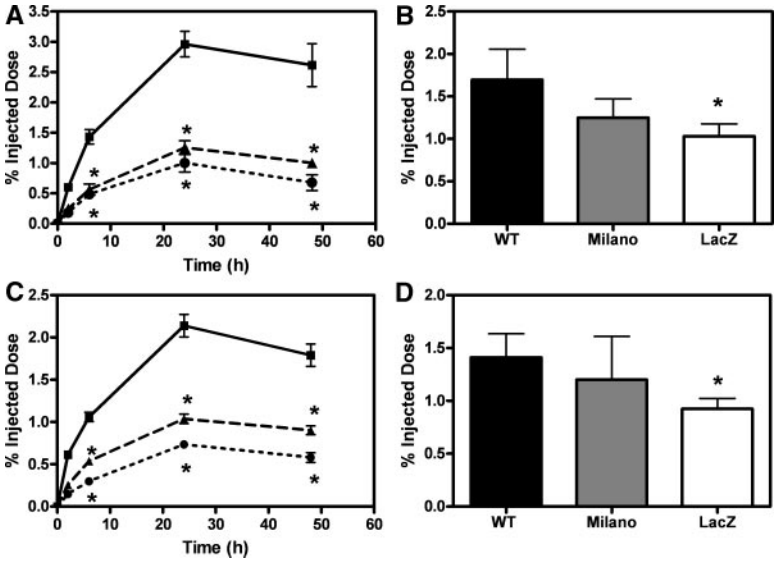


Figure 1. Reverse cholesterol transport assay. Male apoA-I–null mice (n=6 per group) were injected with AAV (1×10^{12} GC) expressing either WT apoA-I, apoA-I_{Milano}, or LacZ 6 weeks before the RCT assay. The mice were then IP injected with [^{3}H]cholesterol-labeled, acLDL-loaded J774 macrophages. Data are expressed as percentage of the [^{3}H]cholesterol tracer relative to total cpm tracer injected ±SD. A, Time course of [^{3}H]cholesterol distribution in plasma. Mice were bled at 2, 6, 24, and 48 hours after injection. B, Fecal [^{3}H]cholesterol tracer levels. Feces were collected continuously from 0 to 48 hours postinjection. C and D, RCT assay was performed as described in A and B except a reduced amount of AAV (2×10^{11} GC) expressing WT apoA-I was used. WT apoA-I (■), apoA-I_M (▲). and LacZ (●). *$P<0.05$ compared to WT.

mice expressing WT apoA-I and similar levels of HDL-C in the WT apoA-I and apoA-I_M mice (Table). Compared with the lacZ control mice, expression of low dose WT apoA-I promoted significantly increased plasma counts and fecal excretion of tracer (Figure 1C and 1D), although to a lesser extent than the higher dose experiment. However, there remained no significant difference between apoA-I_{Milano}–expressing and WT apoA-I–expressing mice in fecal tracer excretion despite the much lower plasma levels of WT apoA-I.

ABCA1-Mediated Cholesterol Efflux

To gain further insight into the effect of apoA-I_M compared with WT apoA-I on RCT, we investigated each of the steps of RCT individually; these steps include ABCA1-mediated cholesterol efflux, the conversion of free cholesterol to cholesteryl ester by LCAT, and SR-BI–mediated selective uptake of cholesterol and cholesteryl ester from HDL particles.[3]

To measure ABCA1-mediated cholesterol efflux, J774 macrophages (the cells used in the macrophage RCT experiments) and MPM were labeled with [^{3}H]-cholesterol. After an equilibration period, 2.5% serum from either WT apoA-I– or apoA-I_M–expressing mice (from RCT #1) was added for 4 hours. As previously reported for MPMs[32] there were no differences in ABCA1-mediated cholesterol efflux (3.6±0.9% for WT apoA-I and 4.6±1.5% for apoA-I_M). This finding was confirmed with J774 macrophages which were used in the macrophage RCT experiments described above (7.9±0.8% for WT apoA-I and 8.5±1.2% for apoA-I_M; Figure 2A). This was a somewhat surprising observation given the lower concentration of apoA-I_M compared to WT apoA-I in the serum samples (Table). To address this issue, recombinant WT apoA-I and apoA-I_M (in the monomeric and homodimeric states) were used at different concentrations (0 to 20 μg/mL) as acceptors for ABCA1-mediated efflux of [^{3}H]-cholesterol from J774 macrophages (Figure 2B). From these experiments, the kinetic constants of the ABCA1 reactions were determined and the K_m values for cholesterol efflux to WT apoA-I, monomeric, and dimeric apoA-I_{Milano} were 4.5±0.5, 2.1±0.3, and 4.8±0.9 μg apoA-I/mL, respectively. The V_{Max} values of the reaction for WT apoA-I, monomeric, and dimeric apoA-I_{Milano} were 7.5±0.3, 5.2±0.2, and 5.8±0.4% cholesterol efflux/4 hour, respectively.

Conversion of Free-Cholesterol to Cholesteryl Ester by LCAT

The second step in the RCT process is the conversion of FC to CE by LCAT in the plasma. The cholesterol esterification

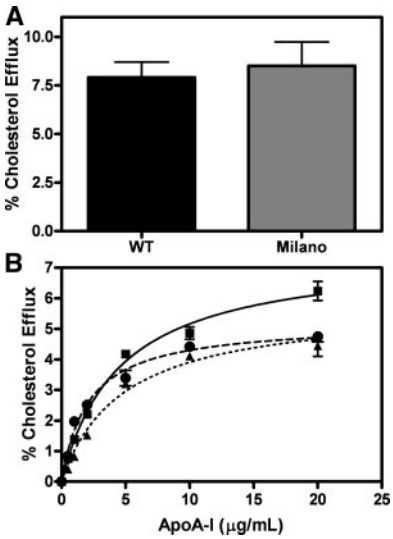


Figure 2. ABCA1-mediated efflux from macrophages to serum. J774 macrophages were labeled with [^{3}H] cholesterol as described under Materials and Methods. These labeled cells were then treated with cAMP overnight to upregulate ABCA1. The lipid efflux was then initiated by the addition of (A) 2.5% serum from either WT apoA-I- or apoA-I_{Milano}-expressing mice or (B) recombinant WT apoA-I or apoA-I_{Milano} (0 to 20 μg/mL). After 4-hour incubation, the medium was removed, filtered, and extracted for lipids and ^{3}H-radioactivity was determined. Curves were generated by fitting the fractional efflux at 4 hours, measured at various concentrations of apoA-I, to the Michaelis-Menten equation. Data represent FC efflux to WT apoA-I (■), monomeric apoA-I_{Milano} (●), and dimeric apoA-I_{Milano} (▲).

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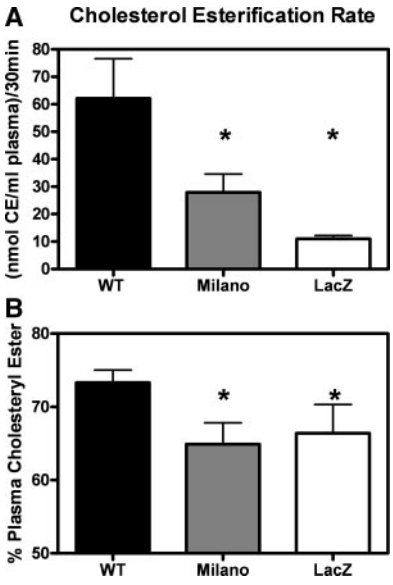


Figure 3. LCAT cholesterol esterification rates. A, Cholesterol esterification rate in the plasma of mice expressing WT apoA-I or apoA-I$_{Milano}$. Whole plasma was radiolabeled with [^{3}H]cholesterol at 4°C overnight and then incubated at 37°C for 30 minutes. LCAT activity was determined as described in Materials and Methods and is expressed as nanomoles of CE formed per milliliter of plasma during the 30 minutes incubation. B, Percentage of cholesterol as CE in the plasma of mice expressing WT apoA-I, apoA-I$_{Milano}$, or LacZ for the mice described in Figure 2A. Values are expressed as mean±SD (n=6 per group). *$P<0.05$ compared to WT.

rates (CER), as determined in the Materials and Methods, for apoA-I$_M$– and LacZ-expressing mice were 45% and 18%, respectively, of that of the WT apoA-I–expressing mice (Figure 3A). Corresponding to the reduction in the CER, plasma CE levels in apoA-I$_M$– and LacZ-expressing mice were significantly reduced compared to the WT apoA-I–expressing animals (Figure 3B).

SR-BI–Mediated Selective Uptake

The third step of the RCT pathway is the SR-BI–mediated selective uptake of cholesterol from HDL in the plasma into the liver. Serum samples from the 48-hour time point of the RCT experiment in Figure 2C, which contain [^{3}H]-cholesterol and [^{3}H]-cholesteryl ester, were diluted in media and applied to Fu5AH rat liver hepatoma cells that express high levels of SR-BI[37] and incubated for 8 hours. After the 8-hour incubation, the cells were processed as described in the Materials and Methods to determine the amount of label that was taken up from the media into the cells (Figure 4). Serum from mice expressing either WT apoA-I or apoA-I$_M$ was equally effective at promoting SR-BI–mediated selective CE and FC uptake into these liver cells.

Discussion

Carriers of the apoA-I$_M$ variant have reduced levels of plasma apoA-I and HDL, but no corresponding increase in cardiovascular disease compared to individuals with the WT apoA-I gene.[10–12] One theory used to explain this inconsistency is that HDL containing apoA-I$_M$ are more efficient at promoting RCT.

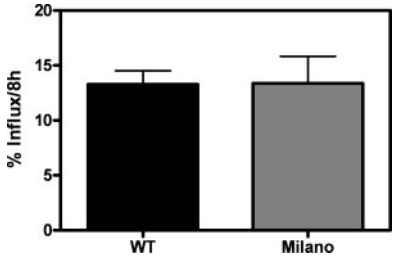


Figure 4. SR-BI–mediated influx of cholesterol from mouse plasma to hepatic cells. [^{3}H]cholesterol and [^{3}H]cholesteryl ester–labeled serum from the 48-hour time point of the experiment described in Figure 2C was diluted to 20% with media and incubated with Fu5AH rat hepatoma cells for 8 hours. Influx was determined as described in Materials and Methods.

This study is the first direct comparison of the ability of WT apoA-I and apoA-I$_M$ to promote macrophage RCT in vivo.

ApoA-I–null mice were injected with AAV expressing either WT apoA-I, apoA-I$_M$, or the LacZ control. The mice were then used in macrophage RCT experiments. In the first macrophage RCT experiment, the relative amounts of apoA-I and HDL-C levels in WT apoA-I– and apoA-I$_M$–expressing mice closely approximated the levels seen in humans,[36] with lower apoA-I and HDL-C levels in mice expressing apoA-I$_M$ (Table). When the mice were assayed for the ability to promote macrophage RCT, there was no significant difference between the WT apoA-I and apoA-I$_M$ groups in fecal excretion of tracer (Figure 1A and 1B) despite the lower HDL-C levels in the latter group. In the second experiment, the amount of WT apoA-I AAV used was reduced so that the resulting HDL-C levels in the WT apoA-I and apoA-I$_M$ mice were equal (Table). Surprisingly, despite much lower plasma apoA-I levels in the mice expressing WT apoA-I, there was still no significant difference in fecal tracer excretion between mice expressing WT apoA-I and mice expressing apoA-I$_M$. This result suggests that apoA-I$_M$ is not more effective than WT apoA-I at promoting macrophage RCT as assessed by this model. To gain greater insight, we dissected the following 3 steps of the RCT pathway, ABCA1-mediated cholesterol efflux, LCAT-mediated conversion of FC to CE, and SR-BI–mediated selective uptake of cholesterol.

Serum from mice expressing apoA-I at levels similar to those listed in Table, RCT 1, was used to measure ABCA1-mediated cholesterol efflux (the transporter responsible for a majority of the cholesterol efflux[37]) from J774 macrophages, the cell type used in the macrophage RCT experiments. Both WT apoA-I and apoA-I$_M$ serum promoted efflux to similar extents (Figure 2A) despite lower levels of apoA-I in the apoA-I$_M$ mice compared to the WT apoA-I mice.[32] To address this issue, we measured the effects of apoA-I concentration on ABCA1-mediated cholesterol efflux from J774 cells (Figure 2B) and determined that K_m of the reactions were in the range of 2 to 5 μg/mL of apoA-I for WT and apoA-I$_M$. In the serum of the WT apoA-I and apoA-I$_M$ mice, if one conservatively estimates that 5% of the circulating apoA-I is in the lipid-free or lipid-poor state,[38,39] the concentrations would be 77 and 61 μg/mL, respectively. These levels are well above the K_m of the ABCA1 reaction so that the reaction would be operating near maximal levels. Even if

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the fact that the levels of apoA-I present in interstitial fluid would be lower than in the serum[40] is taken into account, the concentration of apoA-I would still be well above the K_m of the reaction. The level of lipid-free or lipid-poor apoA-I in the second RCT experiment would also have been well above the saturation point of the ABCA1-mediated reaction. It should also be noted that measurements of pre-β HDL in human apoA-I_M carriers determined that 15% of the apoA-I was in the lipid-free or lipid-poor state,[36] consistent with saturating conditions for the apoA-I/ABCA1 reaction.[41,42] It is also worth noting that although ABCA1 is responsible for transferring cholesterol to lipid-poor apoA-I, ABCG1 can transfer cholesterol to lipidated apoA-I and that the K_m of the ABCG1 reaction is approximately 50 μg/mL HDL protein.[43] The concentration of WT apoA-I and apoA-I_M in the mice is above the K_m of the ABCG1 reaction.

The second step of the RCT process, the conversion of FC to CE by LCAT in the plasma, was measured in plasma using well established methods.[29] The results demonstrated that apoA-I_M was unable to activate mouse LCAT to the same extent as WT apoA-I (Figure 3). This result with plasma HDL confirmed earlier work,[44] showing that recombinant HDL made with dimeric apoA-I_M was 70% less efficient as a substrate for human LCAT than recombinant HDL prepared with WT apoA-I. Structural analysis of the apoA-I_M mutation[45] determined that the loss of the arginine and the gain of the cysteine residue at position 173 destabilized the N-terminal helix bundle of the apoA-I molecule, a region that has been determined to be of particular importance for LCAT activation.[9]

The third step of RCT is the selective uptake of plasma HDL cholesterol by SR-BI into the liver. Serum from WT apoA-I and apoA-I_M mice was equally effective at promoting SR-BI selective uptake (Figure 4). The apoA-I_M mouse serum had a higher FC/CE ratio compared to the WT apoA-I serum (Figure 3B) because of the reduced LCAT activation by apoA-I_M compared to WT apoA-I. Previous work has shown that SR-BI can mediate the selective uptake of both CE and FC from HDL, with the rate constant for the latter being some 60% higher.[46] Thus, the alteration in FC/CE ratio between the WT and apoA-I_M HDL is unlikely to be harmful to net hepatic selective uptake of HDL cholesterol. Furthermore, the K_m for high affinity selective uptake from HDL by SR-BI is approximately 1 μg apoA-I/mL.[47] Thus, the concentrations of apoA-I in the WT apoA-I and apoA-I_M mouse serum samples are far above the K_m that the reaction is saturated and any HDL compositional or quantity differences do not affect SR-BI–mediated selective uptake.

Despite lower apoA-I and HDL-C levels and a reduced ability to activate LCAT, expression of apoA-I_M is as efficient at promoting RCT as expression of WT apoA-I as assessed by this mouse tracer assay. However, even when WT apoA-I is expressed at a lower level (plasma levels one-fourth of those in apoA-I_M mice) to generate similar levels of HDL-C in both groups of mice, expression of apoA-I_M is not more effective in promoting RCT. Thus, if apoA-I_{Milano} does have an enhanced cardioprotective ability, it is likely attributable to other properties of this mutant protein; for example increased antioxidant activity[15] arising from the presence of a cysteine residue in the primary structure of apoA-I or the ability of apoA-I_{Milano} to modulate vascular function.[12,25]

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Disclosures

None.

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Supplement Material

METHODS

***Preparation of ApoA-I Mutants*.** To express human WT apoA-I and apoA-I_{Milano}, the cDNA of interest was cloned into the multiple cloning sites of the pET32a (+) vector, and the target protein was expressed as a histidine-tagged fusion protein with the 109-amino acid thioredoxin (Trx) at the amino terminus (1,2). The resulting plamids were transformed into *Escherichia coli* strain BL21(DE3). These transformed cells were cultured in LB medium at 37˚C and expression of the fusion protein Trx-apoA-I was induced with isopropyl-β-D-thiogalactopyranoside for 3 h. After sonicating the bacterial pellet, the lysate was centrifuged and the supernatant loaded onto a nickel-chelating, histidine-binding resin column (Novagen). The Trx-apoA-I fusion protein bound to the column was eluted, pooled, and dialyzed against 20 mM NH_4HCO_3. Subsequently, the fusion protein was complexed with dimyristoyl phosphatidylcholine (DMPC) to prevent non-specific cleavage, and then cleaved with thrombin to release the Trx. The mixture was then lyophilized, delipidated, and dissolved in 6 M guanidine hydrochloride (GdnHCl) solution. Trx was separated from apoA-I by gel filtration chromatography on a Sephacryl S-300 column. Further purification (>95%) of the proteins was done by gel filtration with Superdex 75 and/or anion exchange chromatography with Q-Sepharose. Purity was confirmed by SDS-PAGE (not shown). Fractions corresponding to dimeric apoA-I_M were pooled separately and confirmed to be >95% dimeric apoA-I_M by SDS-PAGE (not shown). All proteins were stored at -80˚C in lyophilized form and before use were dissolved in the appropriate buffer containing 6M GdnHCL and dialyzed extensively before use.

***Preparation of ApoA-I Adeno-Associated Virus (AAV)*.** WT apoA-I cDNA was mutated using the Quikchange Site-Directed Mutagenesis Kit from Stratagene. Specific primers were designed to mutate the WT apoA-I cDNA in the pAAVmcs plasmid . The

plasmid containing the WT apoA-I cDNA served as the template for the mutagenic PCR reaction. After the PCR reaction the sample was digested with Dpn I, an endonuclease specific for methylated and hemimethylated DNA, to digest the parental DNA template and select for mutation-containing synthesized DNA. The purified mutant plasmid was then transformed into XL1-Blue supercompetent cells. Several colonies were isolated and used to inoculate Luria-Bertani media containing 100mg/ml ampicillin. The resulting plasmids were sequenced and restriction digested to confirm the presence of the intended mutation. The plasmid containing the mutant sequence was then submitted to the University of Pennsylvania Vector Core for use in creating the apoA-I AAV, serotype 8 (3).

Macrophage RCT Studies. RCT studies were performed as previously described (3-8). J774 cells were grown in suspension in RPMI 1640 medium supplemented with 10% fetal bovine serum. Cells were radiolabeled with 5 μCi/mL [^{3}H]cholesterol and cholesterol enriched with 100 μg/mL of acetylated LDL for 48 h. These labeled foam cells were washed twice, equilibrated in medium with 0.2% bovine serum albumin for 6 h, spun down and resuspended in RPMI medium immediately before use.

Experiments were performed in male apoA-I-null mice obtained from Jackson Labs. All mice were fed a standard chow diet. For each experiment, 18 mice (n = 6/group) received intraperitoneal (IP) injections of AAV8-h WT apoA-I (1 X 10^{12} GC), AAV8-h apoA-I_{Milano} (1 X 10^{12} GC) or AAV8-LacZ (1 X 10^{12} GC). On day 42 after vector injection, animal received ip injections of [^{3}H]cholesterol-labeled j774 cells as previously described and were caged individually with unlimited access to food and water. Blood was collected at 2, 6, 24, and 48 hours, and plasmas were used for liquid scintillation counting (LSC). Feces were collected continuously from 0 to 48 h and stored at 4 ˚C before being counted. At 48 hours after injection mice were exsanguinated. The total feces collected were weighed and soaked in Millipore water (1 mL water per 100 mg

feces) overnight at 4 ˚C. An equal volume of ethanol was added the next day and samples were homogenized with 200 µL of the homogenized samples counted in an LSC in duplicate.

Endogenous LCAT Assays. Endogenous LCAT cholesterol esterification rate (CER) in whole plasma was measured using a modified Stokke and Norum procedure (9,10). A tracer amount of [^{3}H]cholesterol in toluene was added into a 50 mg/mL bovine serum albumin (BSA) solution, and the toluene was evaporated under N_2. Seventy five microliters of plasma were incubated with the BSA solution containing 3 X 10^6 dpm of [^{3}H]cholesterol on ice overnight to allow the [^{3}H]cholesterol tracer to equilibrate evenly across the entire spectrum of lipoproteins. The sample was diluted to 300 µL with 0.2 M phosphate buffer (pH 7.4) and divided into three aliquots. Two aliquots were incubated at 37 ˚C for 30 min as duplicates, while the remaining aliquot was maintained at 4 ˚C as a control. All subsequent procedures were the same as described previously (11). Briefly, after incubation, the samples were placed on ice, and the lipid was extracted by the Bligh/Dyer method. The extracted lipids were subjected to TLC using a neutral solvent system to separate the FC and CE radio label which was quantified by scintillation counting. Endogenous LCAT activity was expressed as percentage of FC esterified (calculated as CE/(CE+FC)) and as nanomoles of CE formed per milliliter of plasma (calculated as plasma FC concentration X (%FC esterified/100%) during the 30 min incubation.

Cell Culture. Mouse peritoneal macrophages (MPM) were isolated and prepared as previously described (12). J774 macrophages were grown in RPMI 1640 medium containing 10% FBS + 50 µg/mL gentamycin and incubated at 37 ˚C in a humidified chamber. For efflux experiments, these cells were then seeded in tissue culture plates. The following day wells were washed with MEM-HEPES. The final wash was removed and media containing 1% FBS + [^{3}H]cholesterol or [^{3}H]choline (1 µCi/mL) ± acetylated

LDL (acLDL, 100 μg protein/mL) and + CP-113 818 (2 μg/mL), an ACAT inhibitor, was added for 24 hours. After this, cells were incubated in media containing 0.2% BSA for 18 hours to allow cellular cholesterol radiolabel to equilibrate. In certain experiments, some incubations contained 0.3 mmol/L CTP-cAMP during equilibration. After equilibration, time zero cells were harvested to determine total cpm, protein and FC and CE mass in the cells before adding the acceptors.

Cholesterol Efflux. After [^{3}H]cholesterol labeling the cells, media containing the appropriate acceptor was added for up to 24 hours. In some experiments the ACAT inhibitor CP113 818 (2 μg/mL) was added to the media. To determine the cholesterol efflux, media were sampled at indicated times, filtered, and counted by LSC to determine the [^{3}H] released. [^{3}H] in the media was compared with total [^{3}H] at time zero to determine the percent release of [^{3}H]cholesterol (13).

Data Analysis. Data are from representative experiments and are expressed as mean ± SD. Statistical test for significance was done using an unpaired t-test or 1 way Anova.

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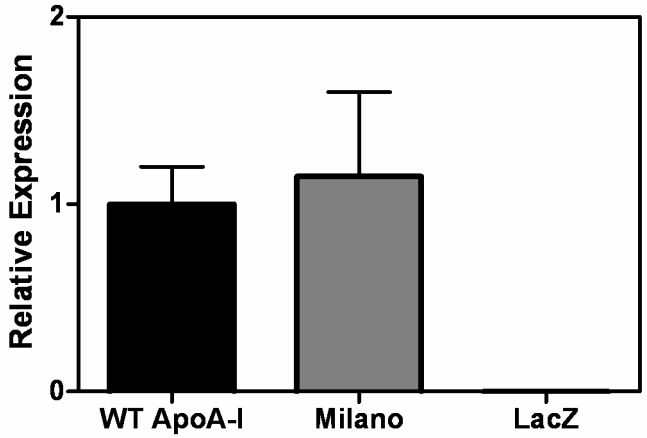
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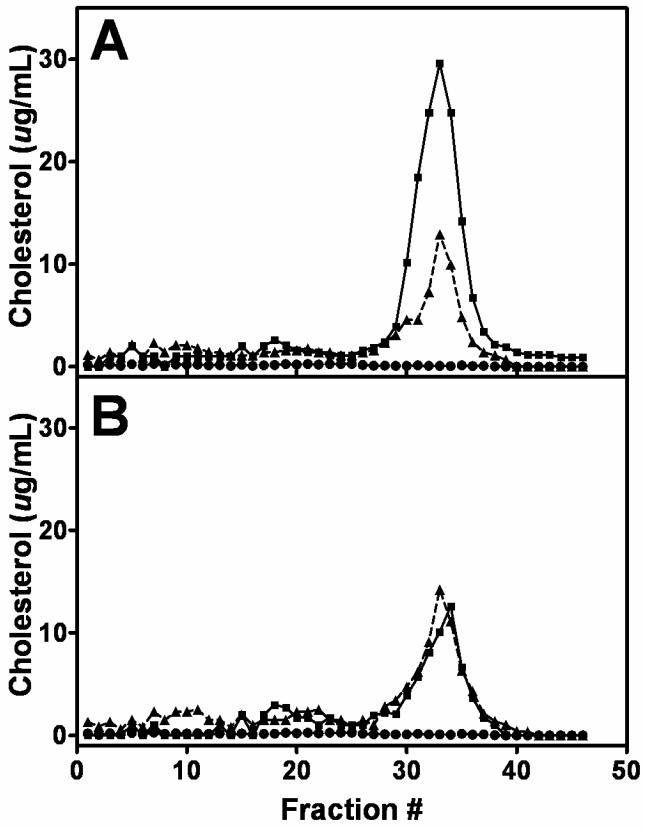
Supplementary Figure I. Quantitative analysis of human apoA-I mRNA from mouse livers 6 weeks post AAV injection. Total RNA was extracted from mouse livers and analyzed by RT PCR. Data are expressed as fold change ± SD and normalized to mouse β-actin.

Supplementary Figure II. FPLC profiles of mouse plasma. Cholesterol mass profile of pooled plasma samples drawn 48 hours after injection of J774 cells. (A) Samples correspond to 48 h time point of Figure 1A, (B) samples correspond to 48 h time point of Figure 1C. WT apoA-I (■), apoA-I_M (▲) and LacZ (●).

Supplementary Figure I



Supplementary Figure II



Supplementary Table I. Analysis of Liver and Bile Samples[*]		
AAV	Liver	Bile
WT[†]	4.0 ± 0.5	0.045 ± 0.017
Milano	3.8 ± 0.5	0.037 ± 0.009
LacZ	3.5 ± 0.2	0.039 ± 0.009

[*] Samples correspond to experiment described in Fig. 2A and 2B

[†] All values expressed as % injected dose

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Eric T. Alexander, Ginny L. Weibel, Michelle R. Joshi, Charulatha Vedhachalam, Margarita de la Llera-Moya, George H. Rothblat, Michael C. Phillips and Daniel J. Rader

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